

OpenAcoustics: An Open-Source Framework for Acoustic Data Capture on Smart Devices and Microcontrollers

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Abstract

Acoustic sensing has become a vital technique across domains such as human-computer interaction, health monitoring, and ubiquitous computing. However, collecting high-quality acoustic data often requires proprietary platforms or specialized hardware, limiting accessibility and reproducibility. We present OpenAcoustics, an open-source system for capturing acoustic data on off-the-shelf smart devices and microcontroller units (MCUs), including smartphones, smartwatches, and embedded controllers. Our system offers real-time audio recording, configurable sampling pipelines, and wireless streaming over BLE UART or Wi-Fi. OpenAcoustics is designed to be lightweight, extensible, and easy to deploy, enabling researchers and developers to collect synchronized, high-fidelity acoustic data in both lab and in-the-wild scenarios. By lowering the barrier to entry for acoustic sensing research, OpenAcoustics fosters reproducibility, rapid prototyping, and broader participation in the exploration of acoustic interaction technologies.

CCS Concepts

• **Human-centered computing** → **Interaction techniques**; *Interaction devices*.

Keywords

Acoustic Sensing, Mobile Devices, Micro-controller Units, Open System

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1 Introduction

Acoustic sensing has emerged as a powerful and versatile modality in a wide range of applications, including activity recognition [7, 8, 10], health monitoring [2, 12], and interaction techniques [4, 6, 13, 14, 16], largely due to the widespread integration of microphones and speakers into smart and IoT devices. Recent advancements in signal processing and AI have enabled systems that

can leverage subtle audio cues to infer fine-grained users' movement (e.g., finger or gaze tracking [6, 9]) or contexts (e.g., object interaction [8]). However, the development and evaluation of such systems often require specialized hardware or proprietary tools, creating significant barriers to entry for researchers and developers.

In this paper, we present OpenAcoustics, an open-source¹ (will be accessible, if accepted), modular acoustic data capture system designed for easy deployment on commercially available smart devices and MCUs. Our system supports real-time recording and transmission of audio signals using built-in microphones on devices such as smartphones, smartwatches, and low-cost MCUs (e.g., XIAO nRF52840 Sense or Adafruit Feather nRF52840 Sense), enabling rapid prototyping and reproducible research in diverse settings. Given the growing body of work that combines acoustic signals with motion data for richer sensing capabilities [11], OpenAcoustics also offers optional IMU data capture for multimodal sensing. By providing a lightweight and extensible framework, we aim to lower the technical barriers to acoustic data collection and system development.

We detail the architecture and implementation of OpenAcoustics across different off-the-shelf platforms and devices, highlighting the following features:

- **High-fidelity audio capture on mobile devices:** Supports 48 kHz, 16-bit PCM audio recording on Android smartphones, which enables stereo recording using two microphones (e.g., top and bottom), and on smartwatches, which support mono recording with a single built-in microphone. Data transmission is performed via Wi-Fi using Transmission Control Protocol (TCP).
- **Low-power audio capture on MCUs:** Enables 16 kHz, 16-bit PCM audio recording on low-cost MCUs with data transmission over Bluetooth Low Energy (BLE) using Universal Asynchronous Receiver/Transmitter (UART).
- **Multimodal sensing support:** Provides optional motion data capture using built-in IMU sensors, allowing synchronized collection with audio signals.
- **Real-time transmission:** Supports both real-time streaming and buffered transmission to mitigate data loss during wireless transfer.



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¹<https://github.com/witlab-kaist/OpenAcoustics>

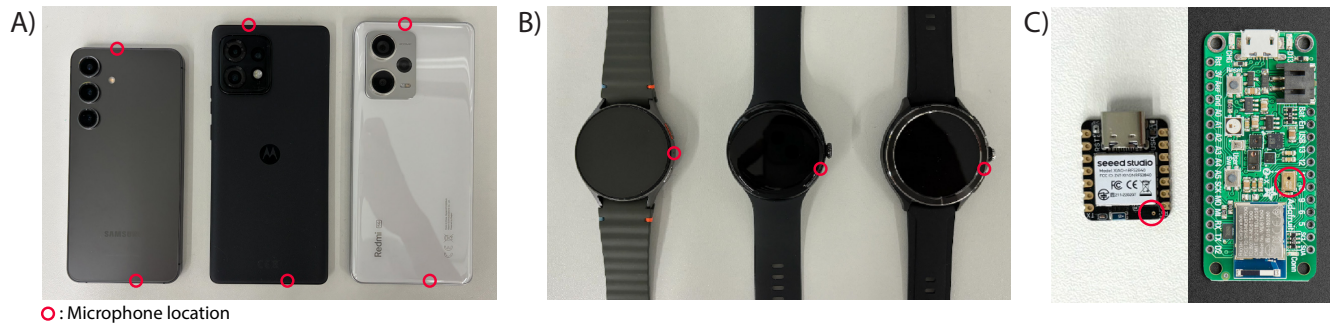


Figure 1: Example platforms supported by *OpenAcoustics*. (A) Smartphones (left to right: Samsung Galaxy S24, Motorola Edge 40, and Xiaomi Redmi Note 12 Pro), (B) Smartwatches (Samsung Galaxy Watch 7, Google Pixel Watch 3, and Xiaomi Watch Pro 2), and (C) MCUs (XIAO nRF52840 Sense and Adafruit Feather nRF52840 Sense)

2 OpenAcoustics

OpenAcoustics is an open-source system for acoustic and multi-modal data collection designed to run on off-the-shelf smart devices and MCUs. The system is compatible with a range of commercially available platforms, including Android-based smartphones and smartwatches, as well as Arduino-compatible MCUs that support audio recording and BLE communication. We validated the system across three smartphones, three smartwatches, and two MCUs (see Figure 1), demonstrating its flexible deployment.

2.1 Architecture

The system follows a modular design, consisting of three key layers: (1) Sensor Interface, (2) Data Processing and Buffering, and (3) Transmission and Storage. Each layer is implemented with flexibility, allowing users to select sample rates, buffer sizes, and streaming modes (real-time or buffered).

2.1.1 Sensor Interface. The sensor interface captures audio signals and motion data from built-in microphones and IMU sensors. Figure 1 illustrates the microphone locations for each tested device. Smartwatches and MCUs are typically equipped with a single microphone, allowing for mono audio capture suitable for simple sensing tasks such as uni-dimensional movement tracking [6] or sound-based activity monitoring [11]. In contrast, smartphones typically include two microphones—positioned at the top and bottom—enabling stereo audio recording. This spatial arrangement can support more advanced tasks such as direction-of-arrival estimation [13, 14]. Motion data is captured reliably at over 50 Hz on smart devices and over 200 Hz on MCUs. On Android devices, sampling rates can be further increased using the *HIGH SAMPLING RATE SENSORS* option [1], subject to hardware capabilities.

2.1.2 Data Processing and Buffering. In the real-time configuration, incoming audio data packets are continuously transmitted to the PC server, immediately decoded from raw bytes into numerical arrays, and processed or streamed with minimal latency. This setup supports applications requiring live monitoring or immediate feedback (e.g., real-time inference).

In the non-real-time configuration, audio data is buffered locally throughout the recording session and processed only after recording ends. This mode allows flexible post-processing and full-session analysis, making it suitable for scenarios where latency is not critical.

2.1.3 Transmission and Storage. For smart devices, *OpenAcoustics* streams audio data via TCP sockets, ensuring reliable, ordered transmission suitable for real-time, high-throughput applications. On average, transmitting a 40 ms audio packet takes 40.004 ms (SD: 7.93 ms), with only 0.004 ms attributed to transmission overhead, and no observed packet loss, demonstrating consistent low-latency performance.

For MCUs, data is transmitted over BLE-UART, offering low power consumption and broad device compatibility. On average, transmitting a full packet of data—247 bytes, containing approximately 7.62 ms of 16-bit audio sampled at 16 kHz—takes 6.24 ms (SD: 0.27 ms) to be received by the server. While sufficient for low-throughput sensing tasks such as IMU data transmission [3], BLE-UART’s lower bandwidth and higher latency can lead to packet loss and discontinuity during high-frequency or large-volume audio streaming.

To address this, we adopted two functions: 1. buffered transmission and 2. synchronized data collection for multi-modal sensing use cases. In buffered transmission, audio data is temporarily stored in the MCU’s RAM for a predefined duration, with BLE transmission occurring after data collection completes. For longer recordings that exceed RAM capacity, an external SD card can be used to store the data more reliably. In synchronized collection, IMU and audio data are transmitted together only when both are ready, reducing the number of BLE transmissions—particularly frequent IMU updates—which are a common cause of packet loss. Under our test conditions (indoor lab environment), asynchronous transmission resulted in a 68.5% audio data capture rate at 16 kHz over 1 second. In contrast, the synchronized mode improved this rate to 98.7%, enabling significantly more reliable real-time data transmission. Although each packet includes only 6-DOF IMU data (3-axis accelerometer and

3-axis gyroscope with a timestamp) once per transmission, *OpenAcoustics* maintains an effective IMU sampling rate of approximately 140 Hz, which is sufficient for typical sensing tasks. [3, 11].

2.2 Extensibility

OpenAcoustics supports multimodal sensing for passive data collection, but recent work has also explored active acoustic sensing by analyzing inaudible sound waves emitted from speakers to detect user gestures or movements [4–6, 8, 14]. While our system does not include built-in audio playback capabilities, users can extend it by integrating open-source active sensing modules such as SonarID [4] (which emits Zadoff–Chu sequences) or SonarSelect [6] (which emits multiple continuous waves).

2.3 Limitations and Considerations

OpenAcoustics currently supports only Android-based devices. Researchers interested in deploying similar systems on iOS platforms may refer to prior open-source work such as LLAP [14].

The system's reliance on Wi-Fi for data transmission on smart devices may pose challenges in field settings with unstable or unavailable networks. In such cases, users are encouraged to modify the system to store audio and motion data directly on the device's internal storage to ensure data integrity.

For microcontroller-based deployments, BLE UART is convenient but limited in bandwidth and susceptible to packet loss during continuous streaming. To improve reliability, users may opt to store data locally on an external SD card, as demonstrated in prior sensing platforms [8, 15].

3 Conclusion

We introduced *OpenAcoustics*, an open-source system for acoustic and multimodal data capture using off-the-shelf smart devices and microcontrollers. By supporting real-time and buffered data collection, synchronized IMU sensing, and flexible transmission, the system enables rapid prototyping and reproducible research across a range of contexts. We hope *OpenAcoustics* lowers the barrier to entry for acoustic sensing studies and serves as a foundation for future research in audio-based interaction.

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